

Molecular Dynamics and Harmonic Heat Capacity of Pristine and Methane-Loaded MOF-5 with a Machine-Learned Interatomic Potential

Author name

August 4, 2026

Abstract

This report develops a simulation workflow for pristine and methane-loaded MOF-5 using the PET-MAD machine-learned interatomic potential. Classical constant-temperature molecular dynamics (MD) is performed with the Atomic Simulation Environment (ASE), while automatic differentiation (AD) through a PET-JAX model provides sparse Cartesian Hessians, normal-mode frequencies, and the quantum harmonic constant-volume heat capacity, C_V . The scientific motivation and reference protocol are taken from Kapil *et al.*, who showed that harmonic quantum lattice dynamics describes empty MOF-5 well, but that loaded MOF-5 requires both anharmonic and nuclear-quantum effects for a quantitative constant-pressure heat capacity, C_P . The present project therefore provides a defensible harmonic route for pristine MOF-5 and a diagnostic local-curvature analysis for MOF-5 containing 100 methane molecules; it does not reproduce the published path-integral C_P . Five short, 2 ps NVT validation trajectories at 100, 200, 300, 400, and 500 K have been completed. They demonstrate execution of the 924-atom loaded system, but are too short to establish equilibration or statistical convergence. This report sets out the theory, implementation, interpretation, and convergence tests required to turn the current validation workflow into a production calculation.

1 Scientific motivation and scope

Metal-organic frameworks (MOFs) combine inorganic nodes with organic linkers to form porous crystalline networks. Their vibrational spectrum contains high-frequency local bond vibrations as well as soft linker, pore, and collective framework modes. Because the contribution of a quantum vibrational mode depends strongly on the ratio of its energy to $k_B T$, a broad frequency distribution produces a strongly temperature-dependent heat capacity. Guest adsorption adds translational, rotational, and host-guest degrees of freedom that may be weakly confined and strongly anharmonic.

The material considered here is MOF-5. Its conventional cubic cell contains 424 framework atoms with composition $C_{192}H_{96}O_{104}Zn_{32}$. In the loaded calculations, 100 methane molecules add 500 atoms, giving a 924-atom system with composition $C_{292}H_{496}O_{104}Zn_{32}$. The principal objectives are:

1. to test whether PET-MAD gives stable finite-temperature dynamics for pristine and methane-loaded MOF-5;
2. to compute PET-MAD force constants efficiently by sparse AD;
3. to obtain and converge quantum harmonic C_V curves;
4. to distinguish results supported by the current implementation from the anharmonic, constant-pressure quantities calculated by Kapil *et al.* [1]; and

5. to establish a reproducible route from short cluster validation jobs to statistically meaningful production calculations.

The workflow has two computational stages. MD samples finite-temperature configurations and diagnoses dynamical stability. The Hessian stage evaluates the local curvature of the potential-energy surface at selected geometries. The latter is not, by itself, an energy-fluctuation estimator and does not become an exact anharmonic heat capacity merely by evaluating more frames.

2 Theory

2.1 Machine-learned interatomic potentials

2.1.1 Learning the potential-energy surface

Within the Born–Oppenheimer approximation, the electronic ground-state energy defines a potential-energy surface (PES) for the nuclei. An interatomic potential replaces an electronic-structure calculation by an approximation to this map,

$$(\mathbf{R}, \mathbf{h}, \mathbf{Z}) \mapsto U(\mathbf{R}, \mathbf{h}, \mathbf{Z}), \quad (1)$$

where $\mathbf{R} = (\mathbf{R}_1, \dots, \mathbf{R}_N)$ are the nuclear coordinates, $\mathbf{h}$ is the periodic cell, and $\mathbf{Z}$ lists the chemical species. A machine-learned interatomic potential (MLIP) represents this map by a differentiable function with parameters θ , fitted to energies and forces from electronic-structure calculations. Modern local MLIPs usually write the extensive energy as

$$U_\theta(\mathbf{R}, \mathbf{h}, \mathbf{Z}) = U_{\text{base}}(\mathbf{Z}) + \sum_{i=1}^N \varepsilon_i(\mathcal{N}_i; \theta), \quad (2)$$

where U_{base} is an optional composition-dependent baseline and $\mathcal{N}_i$ is the learned environment of atom i . The decomposition makes the cost approximately linear in N at fixed density and makes the energy size extensive. The individual atomic energies are a model construction rather than uniquely defined physical observables.

The force field is obtained by differentiating the scalar energy,

$$\mathbf{F}_i = -\frac{\partial U_\theta}{\partial \mathbf{R}_i}, \quad (3)$$

so the predicted forces are conservative, provided the same differentiable energy model is used throughout. Cell derivatives similarly give the virial or stress when the model and its interface support them. Automatic differentiation can be applied a second time to obtain the Cartesian Hessian used later in this project.

2.1.2 From an atomic structure to a graph

A graph neural network (GNN) represents the atoms by vertices and nearby interactions by directed edges. For a periodic system, an edge $(j, \mathbf{n}) \rightarrow i$ is included when

$$\mathbf{r}_{ij\mathbf{n}} = \mathbf{R}_j + \mathbf{h}\mathbf{n} - \mathbf{R}_i, \quad r_{ij\mathbf{n}} = \|\mathbf{r}_{ij\mathbf{n}}\| < r_c, \quad (4)$$

where $\mathbf{n} \in \mathbb{Z}^3$ identifies a periodic image and r_c is a cutoff. This construction removes dependence on the absolute origin. A smooth cutoff function $f_c(r)$, which tends to zero continuously at r_c , prevents discontinuous energy and force changes when a neighbor crosses the cutoff.

Neural networks operate on fixed-width numerical representations. Chemical identity is therefore mapped from a discrete species label to a trainable embedding vector,

$$\mathbf{h}_i^{(0)} = \mathbf{E}_Z[Z_i] \in \mathbb{R}^d, \quad (5)$$

where each row of the embedding matrix $\mathbf{E}_Z$ is learned during training. Geometry may be embedded through radial basis functions,

$$\mathbf{e}_{ij}^{\text{rad}} = f_c(r_{ij})[\phi_1(r_{ij}), \dots, \phi_K(r_{ij})], \quad (6)$$

and, in models that retain directional information, by the displacement $\mathbf{r}_{ij}$, its unit vector, spherical harmonics, or learned functions of these quantities. Here “embedding vector” means a vector in latent feature space; it need not transform as a physical three-dimensional vector.

2.1.3 Message passing and many-body representations

At message-passing layer ℓ , a generic GNN constructs a message on each edge, aggregates the incoming messages in a permutation-independent way, and updates the central representation:

$$\mathbf{m}_{ij}^{(\ell)} = M_\ell(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}), \quad (7)$$

$$\mathbf{m}_i^{(\ell)} = \sum_{j \in \mathcal{N}_i} \mathbf{m}_{ij}^{(\ell)}, \quad (8)$$

$$\mathbf{h}_i^{(\ell+1)} = U_\ell(\mathbf{h}_i^{(\ell)}, \mathbf{m}_i^{(\ell)}). \quad (9)$$

The learned maps M_ℓ and U_ℓ are commonly multilayer perceptrons or attention blocks. Although the graph initially contains only pairwise edges, repeated updates create many-body features: after L layers, an atom can depend on paths extending up to L graph hops. A readout map then converts the final features into the local energies in Eq. (2).

The Point Edge Transformer (PET) used by PET-MAD is a particular GNN that stores a token $\mathbf{f}_{ij}^{(\ell)} \in \mathbb{R}^{d_{\text{PET}}}$ on every directed edge rather than storing information only on atoms [4, 3]. For the environment centered on atom i , PET encodes the neighbor species, relative Cartesian position, and incoming edge message into a token of the schematic form

$$\mathbf{p}_{ij}^{(\ell)} = \sigma(\mathbf{W}_r^{(\ell)} \mathbf{r}_{ij}), \quad (10)$$

$$\mathbf{t}_{ij}^{(\ell)} = \text{MLP}_\ell(\mathbf{f}_{ji}^{(\ell-1)} \parallel \mathbf{E}_Z^{(\ell)}[Z_j] \parallel \mathbf{p}_{ij}^{(\ell)}), \quad (11)$$

where $\parallel$ denotes concatenation and σ is a nonlinear activation. In the first layer the incoming message is absent. A transformer mixes all tokens in one atomic environment by self-attention. For one attention head,

$$\mathbf{q}_a = \mathbf{W}_Q \mathbf{t}_a, \quad \mathbf{k}_b = \mathbf{W}_K \mathbf{t}_b, \quad \mathbf{v}_b = \mathbf{W}_V \mathbf{t}_b, \quad (12)$$

$$\alpha_{ab} = \frac{\exp(\mathbf{q}_a^T \mathbf{k}_b / \sqrt{d_k}) f_c(r_{ib})}{\sum_c \exp(\mathbf{q}_a^T \mathbf{k}_c / \sqrt{d_k}) f_c(r_{ic})}, \quad \mathbf{y}_a = \sum_b \alpha_{ab} \mathbf{v}_b. \quad (13)$$

Multi-head attention repeats this operation in several learned subspaces, followed by feed-forward layers and residual connections. Because no ordering of the neighbors enters Eq. (13), permuting the atoms permutes the edge outputs consistently. PET interprets these outputs as the messages for the next GNN layer and applies learned readout heads across edges and layers to form the total energy.

2.1.4 Physical symmetries

For an isolated non-magnetic system without external fields, a valid scalar energy should be unchanged by translation, rotation or reflection of the whole structure and by permutation of identical atoms:

$$U(\mathbf{R}) = U(\mathbf{R} + \mathbf{a}) = U(\mathbf{Q}\mathbf{R}) = U(\mathbf{P}\mathbf{R}), \quad \mathbf{Q} \in O(3). \quad (14)$$

An invariant scalar is unchanged by a symmetry operation, whereas an equivariant vector changes in the corresponding way; for example, $\mathbf{F}_i(\mathbf{Q}\mathbf{R}) = \mathbf{Q}\mathbf{F}_i(\mathbf{R})$. Translation invariance follows from using relative positions, and the shared local functions and symmetric aggregation enforce permutation invariance. Many MLIPs impose rotational symmetry exactly through invariant geometric descriptors or $E(3)$ -equivariant features. PET instead processes Cartesian displacement vectors with an unconstrained transformer and learns rotational behavior from random rotational augmentation [4, 3]. Thus the PET-MAD model used here is not intrinsically exactly rotation invariant, although its reported rotational discrepancies are small compared with its prediction errors.

2.1.5 Training and domain of validity

Given reference configurations s , a typical energy–force training objective is

$$\mathcal{L}(\theta) = \sum_s \left[w_E \left(\frac{U_\theta^{(s)} - U_{\text{ref}}^{(s)}}{N_s} \right)^2 + \frac{w_F}{3N_s} \sum_{i\alpha} \left(F_{i\alpha,\theta}^{(s)} - F_{i\alpha,\text{ref}}^{(s)} \right)^2 \right], \quad (15)$$

possibly augmented by stress terms and regularization. Because the force labels are derivatives of the PES, each configuration supplies $3N_s$ local slope constraints in addition to its energy. Optimization adjusts the embeddings, message-passing transformations and readout weights to minimize this loss. PET-MAD was trained on a chemically diverse DFT dataset rather than on MOF-5 alone, making it a universal pretrained potential [3].

An MLIP interpolates the reference PES represented by its training data; it does not replace the approximations of the underlying electronic-structure method and need not extrapolate reliably to unfamiliar bonding, charge states, very short distances or thermodynamic phases. Energy and force test errors, trajectory stability and uncertainty indicators are therefore necessary but distinct checks. This project also uses second derivatives, which can be more sensitive than energies or forces and were not directly constrained by the usual loss in Eq. (15). Curvatures and vibrational spectra must therefore be validated explicitly.

2.2 Classical molecular dynamics

2.2.1 Hamiltonian dynamics and statistical averages

Molecular dynamics (MD) generates a trajectory through phase space by propagating the nuclear positions and momenta on a chosen PES. For classical point nuclei with masses m_i , the Hamiltonian is

$$\mathcal{H}(\mathbf{R}, \mathbf{P}) = \sum_{i=1}^N \frac{\mathbf{P}_i^2}{2m_i} + U(\mathbf{R}), \quad (16)$$

and Hamilton’s equations give

$$\dot{\mathbf{R}}_i = \frac{\mathbf{P}_i}{m_i}, \quad \dot{\mathbf{P}}_i = -\frac{\partial U}{\partial \mathbf{R}_i} = \mathbf{F}_i. \quad (17)$$

In exact microcanonical dynamics, $\mathcal{H}$ is conserved. If the dynamics is ergodic and the trajectory is sufficiently long, an equilibrium average can be estimated by a time average,

$$\langle A \rangle = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau A(\mathbf{R}(t), \mathbf{P}(t)) dt. \quad (18)$$

Finite trajectories contain correlated samples, so their statistical precision is controlled by the simulation length relative to the integrated autocorrelation time, not simply by the number of saved frames.

The continuous equations are replaced by a finite-timestep integrator. A representative symplectic scheme is velocity Verlet,

$$\mathbf{P}_i^{n+1/2} = \mathbf{P}_i^n + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^n), \quad (19)$$

$$\mathbf{R}_i^{n+1} = \mathbf{R}_i^n + \Delta t \frac{\mathbf{P}_i^{n+1/2}}{m_i}, \quad (20)$$

$$\mathbf{P}_i^{n+1} = \mathbf{P}_i^{n+1/2} + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^{n+1}). \quad (21)$$

It is time reversible and symplectic, and has global error of order $\mathcal{O}(\Delta t^2)$. It approximately conserves a nearby “shadow” Hamiltonian rather than the exact one. The timestep must resolve the fastest vibration; an apparently controlled temperature does not by itself rule out integration error.

2.2.2 Temperature, ensembles and thermostats

For g unconstrained kinetic degrees of freedom, the instantaneous kinetic temperature is conventionally defined through equipartition,

$$T_{\text{kin}} = \frac{2K}{gk_B}, \quad K = \sum_i \frac{\mathbf{P}_i^2}{2m_i}. \quad (22)$$

Temperature is an ensemble property; T_{kin} fluctuates along a finite canonical trajectory. Constant-energy (NVE) dynamics samples the microcanonical ensemble. To sample the canonical ensemble, whose stationary phase-space density is

$$\rho_{NVT}(\mathbf{R}, \mathbf{P}) \propto \exp[-\beta \mathcal{H}(\mathbf{R}, \mathbf{P})], \quad \beta = (k_B T)^{-1}, \quad (23)$$

the physical variables may be coupled to a thermostat.

For example, Langevin dynamics adds friction and random forces,

$$m_i \ddot{\mathbf{R}}_i = \mathbf{F}_i - m_i \gamma \dot{\mathbf{R}}_i + \boldsymbol{\eta}_i(t), \quad (24)$$

$$\langle \eta_{i\alpha}(t) \eta_{j\beta}(t') \rangle = 2m_i \gamma k_B T \delta_{ij} \delta_{\alpha\beta} \delta(t - t'). \quad (25)$$

The second relation is the fluctuation–dissipation condition that makes the canonical distribution stationary. Weak friction perturbs the natural dynamics less but exchanges heat slowly; strong friction controls temperature rapidly but can overdamp motion. Nosé–Hoover-chain thermostats instead add a sequence of deterministic extended variables and are designed to generate the same canonical distribution while improving ergodicity over a single Nosé–Hoover variable [8]. Thermostat choice affects sampling efficiency and dynamical correlation functions even when equilibrium averages are unchanged.

An isothermal–isobaric simulation additionally evolves the volume or cell so that the target density is proportional to $\exp[-\beta(\mathcal{H} + PV)]$. Flexible-cell algorithms must include the appropriate cell kinetic variables, Jacobian and virial terms; the Martyna–Tobias–Klein scheme is one such extended-system formulation [7]. Reliable NPT dynamics therefore requires a potential with validated cell derivatives or stresses. NVE, NVT and NPT trajectories answer different thermodynamic questions and should not be interchanged solely because their mean temperatures are similar.

2.3 Path-integral molecular dynamics

2.3.1 Quantum statistical mechanics as a ring polymer

Classical MD neglects the wave-like character of the nuclei. For distinguishable quantum nuclei on a single Born–Oppenheimer surface, the Hamiltonian operator and canonical partition function are

$$\hat{\mathcal{H}} = \sum_i \frac{\hat{\mathbf{P}}_i^2}{2m_i} + U(\hat{\mathbf{R}}), \quad Z = \text{Tr} \left[e^{-\beta \hat{\mathcal{H}}} \right]. \quad (26)$$

The kinetic and potential operators do not commute. Splitting the imaginary time $\beta\hbar$ into P slices with a Trotter factorization maps the quantum partition function, in the limit $P \rightarrow \infty$, onto a classical extended system. Its primitive ring-polymer Hamiltonian is

$$Z_P \propto \int d\mathbf{R} d\mathbf{P} e^{-\beta_P H_P}, \quad \beta_P = \frac{\beta}{P}, \quad (27)$$

$$H_P = \sum_{s=1}^P \left[\sum_i \frac{|\mathbf{P}_i^{(s)}|^2}{2m_i} + \sum_i \frac{m_i \omega_P^2}{2} |\mathbf{R}_i^{(s)} - \mathbf{R}_i^{(s+1)}|^2 + U(\mathbf{R}^{(s)}) \right], \quad \omega_P = \frac{P}{\beta\hbar}, \quad (28)$$

with cyclic boundary condition $\mathbf{R}_i^{(P+1)} = \mathbf{R}_i^{(1)}$ [6]. Each physical atom is thus represented by P replicas, or beads, joined by harmonic springs. The physical potential is evaluated independently on every bead, while the springs encode the quantum delocalization of the nuclei in imaginary time. When $P = 1$, Eq. (28) reduces to the classical configurational distribution. Larger P is required as temperature falls or as the maximum physical vibrational frequency increases; convergence must be checked for the chosen isotope masses and temperature range.

Two useful geometrical quantities are the bead centroid and radius of gyration,

$$\bar{\mathbf{R}}_i = \frac{1}{P} \sum_{s=1}^P \mathbf{R}_i^{(s)}, \quad r_{g,i}^2 = \frac{1}{P} \sum_{s=1}^P |\mathbf{R}_i^{(s)} - \bar{\mathbf{R}}_i|^2. \quad (29)$$

The centroid describes the mean imaginary-time position, while the spread of the beads measures nuclear quantum delocalization. For a position-dependent operator $A(\mathbf{R})$, the primitive estimator has the form

$$\langle A \rangle = \lim_{P \rightarrow \infty} \left\langle \frac{1}{P} \sum_{s=1}^P A(\mathbf{R}^{(s)}) \right\rangle_{H_P}. \quad (30)$$

Kinetic energy, pressure and heat capacity require specialized estimators or thermodynamic derivatives; they are not obtained by treating the fictitious ring-polymer kinetic energy as the physical nuclear kinetic energy.

2.3.2 Sampling the ring polymer

PIMD introduces fictitious momenta in order to sample the ring-polymer distribution using MD algorithms. A discrete Fourier or normal-mode transformation diagonalizes the free ring-polymer spring term. Its mode frequencies are

$$\omega_k = 2\omega_P \sin \left(\frac{k\pi}{P} \right), \quad k = 0, \dots, P-1, \quad (31)$$

where $k = 0$ is the centroid mode. The large separation between the slow physical motion and the stiff internal modes makes naive sampling inefficient. Normal-mode or staging coordinates, mass preconditioning, multiple time steps, and mode-specific thermostats address this stiffness. The PILE thermostat applies analytically tuned Langevin friction to the known free ring-polymer modes, while the PILE-L variant also thermostats the centroid [9].

The standard primitive factorization has an asymptotic discretization error of order $\mathcal{O}(P^{-2})$. The fourth-order Suzuki–Chin factorization used in the reference work replaces the bead potential by a bead-dependent effective potential of the general form

$$\tilde{U}_s(\mathbf{R}; \beta) = w_s U(\mathbf{R}) + w_s d_s \left(\frac{\beta \hbar}{P} \right)^2 \sum_i \frac{|\mathbf{F}_i(\mathbf{R})|^2}{m_i}, \quad (32)$$

where w_s and d_s are factorization coefficients that alternate between beads. The force-squared correction accounts for the leading non-commutation of kinetic and potential operators and gives fourth-order convergence for appropriate thermodynamic observables. Direct differentiation would involve Hessians; finite-difference implementations make the scheme practical with general force providers [10]. Ring-polymer contraction can further reduce cost by evaluating slowly varying, long-range components of the potential on fewer contracted beads, while rapidly varying short-range terms remain evaluated on the full polymer.

2.3.3 What PIMD does and does not predict

The trajectories generated by ordinary PIMD are a sampling device for static quantum equilibrium averages. The bead index represents imaginary time, and the artificial bead masses, thermostats and ring-polymer vibrations have no general identification with real nuclear motion. Consequently, ordinary PIMD does not directly yield quantum time-correlation functions, diffusion coefficients or vibrational spectra. Ring-polymer molecular dynamics (RPMD) and centroid molecular dynamics are related but additional approximations for real-time correlation functions; they should not be conflated with PIMD.

PIMD captures nuclear zero-point fluctuations, tunnelling contributions to equilibrium statistics, and anharmonic sampling on the supplied PES, subject to convergence in bead count, timestep, sampling length and system size. It does not correct errors in that PES, include electronic excited states, or automatically change an NVT calculation into a constant-pressure one. For the heat capacity in the reference study, the ring polymer was sampled in a constant-pressure ensemble and C_P was obtained from converged enthalpy derivatives at neighboring temperatures, as discussed below.

2.4 Thermodynamic definitions

2.4.1 Constant-volume and constant-pressure heat capacities

For a system at equilibrium, the constant-volume and constant-pressure heat capacities are

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{V,N}, \quad (33)$$

$$C_P = \left(\frac{\partial \langle H \rangle}{\partial T} \right)_{P,N}, \quad H = E + PV, \quad (34)$$

where E is the total internal energy and H is the enthalpy. For a stable single phase,

$$C_P - C_V = \frac{TV\alpha_V^2}{\kappa_T}, \quad (35)$$

where $\alpha_V = V^{-1}(\partial V/\partial T)_P$ is the volumetric thermal expansion coefficient and $\kappa_T = -V^{-1}(\partial V/\partial P)_T$ is the isothermal compressibility. Equation (35) explains why C_V and C_P need not be interchangeable even when their numerical difference is small for a particular solid.

In an equilibrated classical canonical ensemble, C_V can also be obtained from total-energy fluctuations,

$$C_V = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}. \quad (36)$$

The analogous isothermal–isobaric expression uses enthalpy fluctuations. These relations require a stationary ensemble, long sampling, and uncertainty estimation that accounts for time correlation. Moreover, classical nuclear dynamics gives equipartition rather than the quantum freezing of high-frequency modes. The current code does not use Eq. (36); it calculates quantum harmonic C_V from normal-mode frequencies.

2.4.2 Why classical and quantum predictions differ

In the classical harmonic limit, every quadratic vibrational degree of freedom contributes k_B to C_V , producing the Dulong–Petit limit at all temperatures. Quantum mechanically, a mode of angular frequency ω becomes thermally populated only when $\hbar\omega$ is comparable to or smaller than $k_B T$. High-frequency framework and molecular modes are therefore partially frozen near room temperature. Classical MD can still describe anharmonic structural motion, but it cannot generate the correct quantum populations simply by extending the trajectory.

3 Reference study: Kapil *et al.*

Kapil *et al.* studied empty MOF-5 and cells containing 50, 100, and 150 methane molecules using classical MD, Suzuki–Chin path-integral molecular dynamics (PIMD), and quantum harmonic normal-mode analysis [1]. Their work provides both the physical motivation and an important limit on what the present calculation can claim.

3.1 Force field and starting configurations

The reference work used analytical QuickFF force fields derived from electronic-structure calculations, not PET-MAD [5]. Framework covalent terms were fitted to B3LYP cluster calculations, with MBIS charges obtained from a periodic PBE density. Nonbonded interactions combined Gaussian-distributed electrostatics with an MM3 Buckingham model and a 15 Å cutoff with tail corrections. Methane was inserted with RASPA, configurations with unrealistic contacts were rejected, and canonical Monte Carlo was used before dynamics. Consequently, agreement of timestep and temperature alone would not make the present PET-MAD simulations a numerical reproduction of the paper.

3.2 Published simulation protocols

The associated Nosé–Hoover-chain, constant-pressure, path-integral, and PILE-L methods are described in Refs. [8, 7, 6, 9]. Their production C_P curves were evaluated from neighboring enthalpies,

$$C_P(T) \simeq \frac{\mathcal{H}(T + \Delta T) - \mathcal{H}(T - \Delta T)}{2\Delta T}, \quad \Delta T = 25 \text{ K}, \quad (37)$$

rather than from a single short trajectory. Every temperature point therefore depends on converged, statistically independent simulations at adjacent temperatures.

For the harmonic comparison, optimized structures were used to construct Cartesian Hessians with Yaff and normal modes with TAMkin. Empty MOF-5 was tested with both QuickFF and UFF4MOF. The loaded 100-methane calculation used five different Monte Carlo configurations, each optimized into a local minimum before its harmonic spectrum was evaluated.

3.3 Physical conclusions relevant to this project

The main findings of the reference study are:

Parameter	Classical MD	Suzuki–Chin PIMD
State points	100–500 K, 1 bar	100–500 K, 1 bar
Cell dynamics	Fully flexible cell, zero deviatoric stress	Fluctuating volume with cubic cell shape retained
Integrator	Verlet, 0.5 fs	BAOAB-type multiple-time-step scheme
Thermostat	Three-element Nosé–Hoover chain, 100 fs relaxation	PILE-L ring-polymer thermostat; 100 fs relaxation
Barostat	Martyna–Tobias–Klein, 1000 fs relaxation	Bussi–Zykova–Parrinello form adapted to Suzuki–Chin; 1000 fs relaxation
Quantum discretization	Not applicable	64 beads; long-range forces contracted to 8 beads
Force timesteps	0.5 fs	0.25 fs short range; 1 fs long range
Empty MOF-5 sampling	One 500 ps trajectory	Five independent 125 ps trajectories
Loaded sampling	Five independent 500 ps trajectories	Thirty independent 50 ps trajectories
Discarded equilibration	First 100 ps	First 25 ps
Heat-capacity estimator	Classical thermodynamic sampling	Centered enthalpy derivative with $\Delta T = 25$ K

Table 1: Principal simulation parameters reported by Kapil *et al.* Numerical random seeds and output strides were not reported.

- Empty MOF-5 is quantum mechanical but nearly harmonic over the studied range. Its quantum harmonic C_V is close to the PIMD C_P , with a room-temperature gravimetric value of approximately 0.7 J/g/K.
- Classical MD overestimates the empty-framework heat capacity because it populates all vibrational modes according to equipartition.
- Methane loading introduces strongly anharmonic host–guest motion. The PIMD curves develop a loading-dependent minimum around 200 K, caused mainly by changes in attractive host–guest interactions as confined methane becomes more mobile.
- A harmonic calculation cannot represent diffusion, cage-to-cage motion, or the temperature-dependent localization of methane. Nuclear quantum effects and anharmonic sampling are both needed for a quantitative loaded-system C_P .
- The approximate combination

$$C \approx C_{\text{qn}}^{\text{har}} - C_{\text{cl}}^{\text{har}} + C_{\text{cl}}^{\text{anh}} \quad (38)$$

qualitatively recovers the loaded minimum and becomes useful above roughly 200 K in their 100-methane test. This correction is not implemented in the current repository.

4 Present computational methodology

4.1 Structure preparation and methane insertion

The repository host structure is the conventional cubic MOF-5 cell with 424 atoms and a cell parameter of approximately 25.866 Å. The loaded structure is generated by `utils/insert_methane.py`. Each methane molecule is centered, uniformly randomly rotated using a quaternion, and placed at a uniformly sampled fractional coordinate. Periodic atom–atom distances are checked against all previously placed atoms. The current 100-methane structure used seed 2025, up to 20,000

attempts per molecule, and a minimum distance of 1.5 Å. The structure was validated to contain 924 atoms.

This procedure prevents severe initial overlaps, but it is not an adsorption equilibrium calculation. In particular, it does not reproduce the RASPA insertion and Monte Carlo equilibration used by Kapil *et al.* Several independent insertion seeds and adequate subsequent equilibration are required before drawing conclusions about loading-dependent thermodynamics.

4.2 PET-MAD energy model

The general construction of a local MLIP and the PET architecture are given in Section 2. For the checkpoint used in this project, the model metadata specifies an 8 Å cutoff, three GNN layers, and an adaptive maximum of 40 neighbors. At inference, the learned parameters θ are fixed and Eqs. (2) and (3) are evaluated for each MD configuration.

MD loads the metatomic export `pet-mad-1.5-s_40nn_nostress.pt` through `metatomic-ase`. Harmonic analysis converts the matching checkpoint to a PET-JAX representation. Using the same learned parameters in both paths is essential: a trajectory from one potential must not be differentiated with a different potential. The checkpoint is marked `nostress`; the current workflow therefore keeps the cell fixed and does not assume reliable pressure or cell derivatives. PET-MAD is a universal model rather than a MOF-5-specific fit, so energy, force, and especially curvature accuracy for the relevant structures must be validated [3].

4.3 Classical NVT molecular dynamics

The statistical and dynamical foundations of classical MD are summarized in Section 2.2. In the present workflow, initial velocities are sampled from a Maxwell–Boltzmann distribution at the target temperature and the center-of-mass momentum is removed. The current code does not expose or record a velocity seed, so exact dynamical replicas are not yet reproducible.

Temperature is controlled with ASE’s Langevin integrator [2], using $\gamma = 0.01 \text{ fs}^{-1}$ in Eq. (25). The friction time $\gamma^{-1} = 100 \text{ fs}$ happens to equal the thermostat relaxation time used in the reference study, although Langevin and Nosé–Hoover-chain dynamics are not equivalent.

The completed loaded validation configurations use 4,000 steps with a 0.5 fs timestep, for 2 ps at each of 100, 200, 300, 400, and 500 K. Positions, cell, energies, and momenta are written every step, giving 4,001 trajectory frames including the initial configuration. The fixed cell and NVT thermostat mean that pressure, thermal expansion, and C_P are outside the present MD model.

4.4 Harmonic approximation and Cartesian Hessian

Let $\mathbf{R}_0$ be a stationary reference structure and $\mathbf{u} = \mathbf{R} - \mathbf{R}_0$. Expanding the potential energy gives

$$U(\mathbf{R}_0 + \mathbf{u}) = U(\mathbf{R}_0) + \underbrace{\nabla U(\mathbf{R}_0) \cdot \mathbf{u}}_{=0 \text{ at a minimum}} + \frac{1}{2} \mathbf{u}^T \mathbf{H} \mathbf{u} + \mathcal{O}(u^3), \quad (39)$$

with Cartesian Hessian

$$H_{i\alpha,j\beta} = \frac{\partial^2 U}{\partial R_{i\alpha} \partial R_{j\beta}} = -\frac{\partial F_{i\alpha}}{\partial R_{j\beta}}. \quad (40)$$

For N atoms, $\mathbf{H}$ has dimensions $3N \times 3N$; for the 100-methane cell this is 2772×2772 .

PET-JAX differentiates the model computation graph rather than using displaced finite differences. The implementation uses forward-over-reverse Hessian–vector products,

$$\mathbf{H} \mathbf{v} = \left. \frac{d}{d\epsilon} \nabla U(\mathbf{R} + \epsilon \mathbf{v}) \right|_{\epsilon=0}. \quad (41)$$

Because a local message-passing model couples only graph-connected atoms, the Hessian has a block-sparse structure. SADMOf [11] constructs a graph-derived sparsity pattern and ASDEX colors its columns so that several independent columns can be reconstructed from one differentiated evaluation. The current settings are single precision, graph depth `hops=3`, HVP chunk size 1, and rematerialization enabled. These are computational choices, not established convergence values. In particular, the graph-hop pattern must be enlarged until frequencies and C_V stop changing, and should be checked against a dense or less aggressively truncated reference where feasible.

4.5 Normal modes and quantum harmonic heat capacity

The Hessian is symmetrized and mass weighted to obtain

$$D_{i\alpha,j\beta} = \frac{H_{i\alpha,j\beta}}{\sqrt{m_i m_j}}, \quad D\mathbf{e}_k = \omega_k^2 \mathbf{e}_k. \quad (42)$$

The code enforces the force-constant acoustic sum rule before diagonalization,

$$\sum_j H_{i\alpha,j\beta} = 0, \quad (43)$$

so that uniform translations have zero restoring force. The current SADMOf observable maps negative eigenvalues to zero frequency and excludes modes below 10^{-3} cm^{-1} . Consequently, the stored frequency array does not retain the sign or magnitude of imaginary frequencies. Stability must instead be checked from a signed-eigenvalue analysis or an independently validated normal-mode tool before interpreting a thermally displaced or unrelaxed frame.

For each retained mode, define

$$x_k = \frac{\hbar\omega_k}{k_B T} = \frac{hc\tilde{\nu}_k}{k_B T}, \quad (44)$$

where $\tilde{\nu}_k$ is the wavenumber. The quantum harmonic contribution is

$$C_{V,k} = k_B \left[\frac{x_k/2}{\sinh(x_k/2)} \right]^2, \quad (45)$$

and the gravimetric result is

$$c_V(T) = \frac{N_A}{M_{\text{cell}}} \sum_{k \in \mathcal{V}} C_{V,k}, \quad (46)$$

in J/g/K. The mass M_{cell} includes methane for a loaded system, so both the spectrum and the normalization change with loading.

One Hessian produces the entire configured 100–500 K curve in 10 K increments. The temperature appearing in Eq. (46) controls oscillator populations; it does not require a separate MD trajectory at every analysis temperature when the force constants are assumed temperature independent.

4.6 Role of trajectory frames and averaging

The program reads saved ASE trajectories only to obtain geometries. It computes one Hessian for each explicitly selected frame, serially on one JAX device, and stores a separate C_V curve for every frame. It does not compute a Hessian at every MD step unless every frame is explicitly requested.

For a conventional harmonic calculation, the scientifically preferred reference is an energy-minimized geometry and one Hessian is sufficient. For a loaded system with several local minima,

multiple independently initialized and optimized configurations can characterize basin-to-basin sensitivity, as in the reference paper. Hessians evaluated directly at finite-temperature MD frames describe instantaneous local curvature. If N_f equilibrated, well-separated frames are used, their diagnostic mean is

$$\overline{c_V}(T) = \frac{1}{N_f} \sum_{s=1}^{N_f} c_V^{(s)}(T), \quad (47)$$

with sample variance

$$s_{c_V}^2(T) = \frac{1}{N_f - 1} \sum_{s=1}^{N_f} \left[c_V^{(s)}(T) - \overline{c_V}(T) \right]^2. \quad (48)$$

Adjacent frames are correlated. A standard error should therefore use an effective sample count, approximately

$$N_{\text{eff}} \simeq \frac{N_f}{2\tau_{\text{int}}/\Delta t_{\text{sample}}}, \quad \text{SE}[\overline{c_V}] \simeq \sqrt{\frac{s_{c_V}^2}{N_{\text{eff}}}}, \quad (49)$$

or a block/bootstrap analysis over independent trajectories.

Equation (47) is an ensemble average of local harmonic models, not a formally exact anharmonic partition function. It may reveal temperature-dependent softening and configurational variability, but it does not include barrier crossing, methane diffusion, or the correct free-energy weights of vibrational basins automatically. It must therefore be labelled a local-harmonic or effective-spectrum diagnostic, especially for loaded MOF-5.

5 Completed validation campaign

Five independent Slurm jobs were run for the same 100-methane starting structure. The log files show that all jobs reached their requested 4,000 MD steps and produced 4,001 frames. Table 2 reports only directly observed endpoint information; it is not an equilibration analysis.

Target T (K)	Initial T (K)	Final T (K)	Duration (ps)	Frames
100	95.4	99.2	2.0	4001
200	215.7	207.0	2.0	4001
300	296.3	300.7	2.0	4001
400	400.3	399.4	2.0	4001
500	484.2	495.5	2.0	4001

Table 2: Completed short NVT validation trajectories for MOF-5 plus 100 methane molecules. Temperatures are instantaneous values from the first and final log records.

The final temperatures are close to their targets, and successful completion shows that the structure, model export, CUDA calculator, neighbor-list backend, thermostat, and trajectory output operate together for 924 atoms. However, 2 ps is only 20 thermostat relaxation times and is far shorter than the 25–100 ps discarded equilibration intervals in the reference work. Endpoint agreement with the thermostat does not demonstrate stationary potential energy, equilibrated methane positions, structural stability, or decorrelated sampling.

The heat-capacity submission script currently selects final frame 4000 and writes a unique NPZ result for each MD temperature. This is an appropriate resource and integration diagnostic. At the time of writing, no `heat-capacity*.npz` files are present in the workspace, so no numerical PET-MAD C_V values are reported here.

6 Recommended production and analysis protocol

6.1 Pristine MOF-5: primary quantitative target

The strongest first scientific calculation is the pristine-framework quantum harmonic C_V , because this is the regime in which Kapil *et al.* found close agreement between harmonic and path-integral results.

1. Relax the atomic coordinates with the same PET-MAD checkpoint used for the Hessian. Report maximum residual force and do not relax the cell unless a stress-capable, validated model is adopted.
2. Check stability using signed eigenvalues and inspect low-frequency modes. A large number of unstable modes invalidates the harmonic interpretation.
3. Converge float32 against float64, graph hops, HVP chunking, acoustic sum-rule treatment, and frequency threshold.
4. Converge the conventional-cell result with respect to supercell size or phonon wave-vector sampling. A single periodic cell Hessian samples only the corresponding finite set of modes.
5. Compare the curve and the room-temperature value with the approximately 0.7 J/g/K reference scale, while stating that PET-MAD replaces QuickFF and that model error remains to be quantified.

6.2 Loaded MOF-5: staged diagnostic programme

For the 100-methane system, a sensible progression is:

1. Use the completed final-frame Hessian only to benchmark runtime, memory, and numerical stability.
2. Extend MD until temperature, potential energy, framework descriptors, and methane distributions are stationary. Discard equilibration before selecting frames.
3. Estimate autocorrelation times for energy and representative host-guest observables, then select well-separated production frames.
4. Prefer separate Hessian jobs for independent frames because the current implementation is serial and single-GPU. Start with a small number, increase it until the mean curve and uncertainty stabilize, and retain every individual curve.
5. Repeat the whole workflow for multiple methane insertion and velocity seeds. Basin-to-basin variation cannot be measured from one initial structure.
6. Label the average as a trajectory-frame harmonic diagnostic. Do not identify a minimum near 200 K as the paper’s physical C_P minimum without an anharmonic, quantum, constant-pressure method.

An exact reproduction of the loaded reference calculation would additionally require a validated pressure/stress model, cell dynamics, the published or an equivalently validated host-guest potential, Suzuki-Chin PIMD, ring-polymer contraction, multiple timesteps, controlled replicas and seeds, enthalpy finite differences at $T \pm 25$ K, restart support, and uncertainty propagation across neighboring temperatures.

6.3 Equilibration and convergence criteria

Production decisions should be based on observables, not a fixed waiting time alone. At minimum, inspect:

- running and block averages of temperature, potential energy, and total energy;
- maximum and distribution of force magnitudes;
- framework bond integrity, pore geometry, and cell consistency;
- methane center-of-mass distributions, host–guest distances, and mean-squared displacements;
- sensitivity to timestep and thermostat friction;
- agreement among independent seeds; and
- convergence of C_V with frame count, frame spacing, precision, sparsity depth, cell size, and mode treatment.

The timestep should first be validated in short NVE tests by checking energy drift. Thermostat convergence should then be tested separately, because a thermostat can conceal an integration error while maintaining the desired kinetic temperature.

7 Implementation, outputs, and reproducibility

The active MD path is `run.py` with TOML configuration files. Methane insertion is handled by `utils/insert_methane.py`; harmonic analysis is handled by `utils/heat_capacity.py`. The latter stores the run name, structure, trajectory, PET-JAX checkpoint, selected frame indices, temperature grid, frequency arrays, gravimetric C_V curves, precision, graph hops, HVP chunk size, and rematerialization choice in an NPZ file.

Cluster calculations use one node, one task, and one GPU because neither the MD nor Hessian code is MPI or multi-GPU parallel. Independent temperature, replica, or Hessian-frame jobs are the available level of parallelism. Slurm job identifiers and resource use should be retained; elapsed time and peak memory can be recovered with, for example,

```
sacct -j JOB_ID --format=JobID,State,Elapsed,MaxRSS,AllocTRES
```

Wall time and memory requests should be based on representative jobs rather than extrapolated from the short MD runtime. Separate output directories and prefixes are required for every loading, temperature, seed, and replica.

A reproducible result should record:

- source structure, cell, atom count, chemical composition, loading, and methane-insertion seed and rejection criteria;
- PET-MAD checkpoint hash, metatomic export, PET-JAX conversion, software environment, device, and numerical precision;
- ensemble, target temperature, timestep, thermostat and friction, number of steps, output stride, velocity seed, equilibration interval, production interval, and restart history;
- Hessian frame indices, relaxation state, residual forces, graph hops, HVP chunk size, rematerialization, acoustic sum rule, frequency threshold, and temperature grid; and
- independent-replica count, autocorrelation or block length, uncertainty estimator, and all convergence tests.

8 Limitations and interpretation

The principal limitations of the current results are:

- PET-MAD and QuickFF describe different potential-energy surfaces; agreement with the paper cannot be assumed.
- The `nostress` checkpoint and fixed cell preclude the paper’s constant-pressure cell fluctuations and thermal expansion.
- Classical nuclei are used in MD; nuclear quantum effects enter only through the harmonic oscillator formula.
- The 2-ps loaded trajectories are validation runs, not equilibrated production samples.
- The random insertion is not an adsorption or Monte Carlo equilibrium distribution, and MD velocity seeds are not controlled.
- A Hessian at a thermally displaced frame is not necessarily evaluated at a minimum. Imaginary directions and basin crossing complicate its physical interpretation.
- Averaging local harmonic frame curves is not an exact anharmonic C_V and cannot recover the published loaded-system C_P minimum by construction.
- Finite-size, sparse-Hessian, precision, model-validity, and statistical errors have not yet been quantified.

These qualifications do not make the current calculations uninformative. The workflow is valuable for checking model stability, measuring computational cost, comparing local spectra, and establishing whether a converged pristine harmonic calculation is feasible. They define the boundary between a useful diagnostic and a thermodynamic claim.

9 Conclusions

The repository now implements a complete technical path from periodic MOF-5 structures and methane insertion to GPU NVT trajectories and sparse-AD harmonic C_V curves. The completed 100–500 K loaded trajectories confirm that the 924-atom workflow executes successfully, but their 2 ps duration supports validation only. The most defensible first quantitative target is a relaxed and converged harmonic C_V for pristine MOF-5. Multiple optimized configurations or equilibrated trajectory frames can then be used to assess variability in loaded MOF-5, with an explicit local-harmonic label and statistical uncertainty. Quantitative reproduction of Kapil *et al.*’s loaded C_P requires a separate anharmonic, constant-pressure, nuclear-quantum implementation.

References

- [1] V. Kapil *et al.*, *Modeling the Structural and Thermal Properties of Loaded Metal–Organic Frameworks: An Interplay of Quantum and Anharmonic Fluctuations*, *Journal of Chemical Theory and Computation* **15**, 3237–3249 (2019), doi:[10.1021/acs.jctc.8b01297](https://doi.org/10.1021/acs.jctc.8b01297).
- [2] A. H. Larsen *et al.*, *The Atomic Simulation Environment—a Python Library for Working with Atoms*, *Journal of Physics: Condensed Matter* **29**, 273002 (2017), doi:[10.1088/1361-648X/aa680e](https://doi.org/10.1088/1361-648X/aa680e).

- [3] A. Mazitov *et al.*, *PET-MAD as a Lightweight Universal Interatomic Potential for Advanced Materials Modeling*, Nature Communications **16**, 10653 (2025), [doi:10.1038/s41467-025-65662-7](https://doi.org/10.1038/s41467-025-65662-7).
- [4] S. N. Pozdnyakov and M. Ceriotti, *Smooth, Exact Rotational Symmetrization for Deep Learning on Point Clouds*, Advances in Neural Information Processing Systems **36** (2023), [conference paper](#).
- [5] L. Vanduyfhuys *et al.*, *QuickFF: A Program for a Quick and Easy Derivation of Force Fields for Metal–Organic Frameworks from Ab Initio Input*, Journal of Computational Chemistry **36**, 1015–1027 (2015).
- [6] M. Ceriotti, J. More, and D. E. Manolopoulos, *i-PI: A Python Interface for Ab Initio Path Integral Molecular Dynamics Simulations*, Computer Physics Communications **185**, 1019–1026 (2014).
- [7] G. J. Martyna, D. J. Tobias, and M. L. Klein, *Constant Pressure Molecular Dynamics Algorithms*, Journal of Chemical Physics **101**, 4177–4189 (1994).
- [8] G. J. Martyna, M. L. Klein, and M. Tuckerman, *Nosé–Hoover Chains: The Canonical Ensemble via Continuous Dynamics*, Journal of Chemical Physics **97**, 2635–2643 (1992).
- [9] M. Ceriotti, M. Parrinello, T. E. Markland, and D. E. Manolopoulos, *Efficient Stochastic Thermostatting of Path Integral Molecular Dynamics*, Journal of Chemical Physics **133**, 124104 (2010).
- [10] V. Kapil, J. Behler, and M. Ceriotti, *High Order Path Integrals Made Easy*, Journal of Chemical Physics **145**, 234103 (2016), [doi:10.1063/1.4971438](https://doi.org/10.1063/1.4971438).
- [11] M. F. Langer, M. Ceriotti, and collaborators, *SADMOF: Sparse Automatic Differentiation for Heat Capacity of Metal–Organic Frameworks*, project software and methodology under development (version used by this repository, 2026).